Infinite NPA depth, finite semidefinite lift

An exact quantum region in the simplest Bell experiment

The quantum region of the CHSH value and the sum of two marginal averages requires infinite Navascués–Pironio–Acín depth, yet admits an exact finite semidefinite lift. We construct this lift by showing that every supporting bound is attained with equal measurement overlaps: one angle determines the entire boundary. Finite NPA certificates must remain valid for two independently varying angles, where the obstruction to finite depth survives. The same region has the smallest semidefinite extension degree possible for a nonpolyhedral set, separating infinite hierarchy depth from minimal semidefinite complexity.

Alice and Bob choose between two measurements with outcomes ± 1. Write $a_x = \langle A_x \rangle$, $b_y = \langle B_y \rangle$ and $E_{xy} = \langle A_x B_y \rangle$. Retain

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \tag{1}$$

Their quantum region $\mathcal{W} = \{(M, S) : p \in \mathcal{Q}_{222}\}$ includes all quantum behaviors, with no equality imposed on the marginal averages. For its level-L NPA outer approximation $\mathcal{W}_L$, the preceding Letter [1] and the construction below give

$$\boxed{\begin{array}{c} \mathcal{W} \subsetneq \mathcal{W}_L \quad \text{for every finite } L, \\ \operatorname{sxdeg}(\mathcal{W}) = 2. \end{array}} \tag{2}$$

We construct one lift with positive semidefinite (PSD) blocks of orders five and four. Extension degree is the smallest possible largest block order among all finite lifts; its value two follows separately from a planar representation theorem.

Why one angle suffices. To determine $\mathcal{W}$, consider an arbitrary supporting inequality $hS + \alpha M \le \lambda$, with real h, α. Jordan's lemma reduces its validity to positivity of two-qubit slacks $\lambda I - H$, where

$$H = h\,\mathrm{CHSH} + \alpha(ZI + IZ),$$
$$A_1 = cZ + \sqrt{1-c^2}X, \quad B_1 = dZ + \sqrt{1-d^2}X.$$

Here $A_0 = B_0 = Z$ and $c, d \in [-1, 1]$ are the cosines of the local measurement angles. Deterministic strategies attain $\ell = 2|h| + 2|\alpha|$, so $\lambda \ge \ell$ is necessary. Above this bound, the first three elementary symmetric functions of the eigenvalues of $\lambda I - H$ are nonnegative. Its determinant therefore decides positivity: if it too is nonnegative, the characteristic polynomial of $H - \lambda I$ has nonnegative coefficients and hence no positive root.

Put $x = (1-c)/2$, $y = (1-d)/2$, and write $D(x, y) = \det(\lambda I - H)$. The determinant takes the form

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \tag{3}$$

where F depends on the product alone. In the nonclassical range $|\alpha| \le |h|$, the coefficient of $x + y$ is nonnegative. At fixed product, $x + y \ge 2\sqrt{xy}$, with equality at $x = y = \sqrt{xy}$. Thus equal overlaps minimize the determinant; positivity there implies positivity for every pair. For $|\alpha| \ge |h|$, the quantum support is the deterministic value ℓ: average the two signed one-sided tilted-CHSH bounds and add the remaining marginal slack. Equal overlaps attain this value too. Hence they suffice for *every* support direction.

At equal overlaps $c = d$, the singlet decouples. In the remaining three-dimensional block, set $c = (1 - t^2)/(1 + t^2)$ and multiply the slack by $(1 + t^2)^2$. With

$$p_0 = (1+t^2)^2, \qquad p_1 = 1-t^4, \qquad p_2 = 1+4t^2-t^4,$$

the result is the quartic matrix

$$K(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2\alpha p_0 & 0 \\ -2\alpha p_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}. \tag{4}$$

Its value at zero and leading coefficient force $\lambda \ge \ell$, which also controls the singlet. The missing endpoint follows by continuity. We have therefore proved

$$\max_{\mathcal{W}}(hS + \alpha M) \le \lambda \quad \Longleftrightarrow \quad K(t) \succeq 0 \quad \forall t \in \mathbb{R}. \tag{5}$$

All three support coefficients still enter *affinely*. This is what will give one lift for the whole region.

From one variable to one finite lift. A PSD real univariate matrix polynomial is a polynomial matrix sum of squares, including on its singular boundary [2]. For a quartic, the factors have degree at most two. For $\xi \in \mathbb{R}^3$, average the Gram matrix under $(t, \xi) \mapsto (-t, \mathrm{diag}(1, 1, -1)\xi)$. Mixed parities cancel, splitting the nine quadratic features into

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T,$$
$$Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Consequently Eq. (5) is equivalent to the polynomial identity

$$\xi^T K(t) \xi = Z_+^T G Z_+ + Z_-^T J Z_-, \qquad G \succeq 0, \quad J \succeq 0, \tag{6}$$

with G of order five and J of order four. Matching coefficients gives sixteen affine equations.

The adjoint coefficient maps give the membership test

$$\begin{aligned}\mathcal{W} = \{(m(z), s(z)) : z \in \mathbb{R}^{16}, \\ \ell(z) = 1,\ U(z), V(z) \succeq 0\}.\end{aligned} \tag{7}$$

The matrices U, V are the adjoints of the coefficient map, of orders five and four; ℓ, m, s are its linear coordinate maps. Their rational entries are given in the complete proofs.

There is a direct physical proof of the reverse inclusion. For a real unit triplet state ξ at angle t, choose

$$U = \frac{Z_+ Z_+^T}{(1+t^2)^2}, \qquad V = \frac{Z_- Z_-^T}{(1+t^2)^2}.$$

These matrices obey the shared-entry constraints, normalize to one, and return the state's actual M, S. Their limits include $t = \infty$. The triplet images have every quantum support value; the omitted singlet points $(0, -2c)$ are mixtures of triplet endpoints $(0, \pm 2)$. Their convex hull is therefore $\mathcal{W}$. Mixing witnesses admits every point. Conversely, pairing any feasible matrices with every Gram certificate gives $\lambda - hS - \alpha M \geq 0$, so no point outside $\mathcal{W}$ is admitted.

The compact triplet image is connected. A point in its planar convex hull lies on a chord joining two image points: otherwise its connected radial image would lie in an open semicircle, giving a separating line. Thus every point of $\mathcal{W}$ needs at most two real equal-overlap strategies, with both membership matrices of rank at most two.

Why the NPA obstruction remains. Equal overlaps determine support values; a finite NPA certificate must represent the bound for independently chosen measurement angles. The earlier nontermination proof [1] uses this freedom: near the local threshold, an NPA error bounded by a constant times the quantum violation above the local bound would force a shifted Motzkin polynomial to be a sum of polynomial squares. The complete earlier argument is retained in the companion.

Minimal block order and thirty-two exact projections. For $f(\alpha) = \max_{\mathcal{W}}(S + \alpha M)$, the exact quantum bound has the expansion [1]

$$f(1-T) = 4 - 2T + \tfrac{4}{3}T^3 + O(T^4), \qquad T \downarrow 0. \tag{8}$$

A polygon would have affine support sufficiently close to $\alpha = 1$, contrary to Eq. (8). Thus $\mathcal{W}$ is nonpolyhedral. It is also closed, convex and semialgebraic, so Scheiderer's planar theorem gives $\operatorname{sxdeg}(\mathcal{W}) \leq 2$ [3]. Order-one lifts are polyhedral, hence $\operatorname{sxdeg}(\mathcal{W}) = 2$. The blocks $5, 4$ are explicit; the optimal-order lift is existential.

Physical input and output relabellings give the same quantum region for

$$\left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij}\right), \qquad \prod_{ij} \chi_{ij} = -1.$$

The eight CHSH sign patterns and four selected input pairs give 32 distinct planes. Intersecting their fixed lifts with any NPA relaxation makes every support value on these planes exact, including all the directions responsible for finite-level nontermination.

References

[1] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, manuscript (2026). Its complete nontermination proof is given in the companion, Part II.

[2] C. Hanselka and R. Sinn, *Math. Z.* **292**, 83–101 (2019), Corollary 2.8; arXiv:1707.08261.

[3] C. Scheiderer, *SIAM J. Appl. Algebra Geom.* **5**, 114–139 (2021), Theorem 0.1; arXiv:2004.04196.